

KAGISO MOSEHLA

Honours Computer Science Student Specializing in Artificial Intelligence

 Johannesburg, South Africa

 (+27) 72-938-5320

 kagisoformal228@gmail.com

 Portfolio

EXPERTISE

Programming (SAS, R, SQL, Java
Python)
Big Data (Hadoop, Hive)
Data Processing (NumPy, Pandas)
Data Visualization & Reporting
(Power BI, Excel)
Machine Learning & AI (scikit-learn,
TensorFlow, NLP, LLM, Deep
Learning)
Software & Tools (Git, APIs,
Networking)
Soft Skills (Analytical Thinking &
Problem Solving,
Understanding & Decision,
Support, Stakeholder
Communication, Teamwork,
Time Management & Delivery
Focus)

EDUCATION

BSC HONS IN COMPUTER
SCIENCE(AI)
University of Pretoria
2026 - Present

BSC MATHEMATICAL STATISTICS
AND COMPUTER SCIENCE
University of Cape Town
2021 – 2024

CAREER SUMMARY

Honours Computer Science student specializing in Artificial Intelligence, with strong foundations in machine learning, data analysis, and software development. Experienced in building predictive models, engineering datasets, and optimizing model performance on real-world financial data.

Currently working on NLP and LLM-based projects, with a strong interest in AI-driven automation, intelligent systems, and scalable solutions. Skilled in Python, SQL, and TensorFlow, with the ability to translate complex data into actionable business insights

PROFESSIONAL EXPERIENCE

JUNIOR SPECIALIST – REGULATORY INFORMATION MANAGEMENT NOV 2025 – JAN 2026
Absa Group Limited

- Respond to regulatory requests, subpoenas, and internal investigations by extracting and analyzing large-scale financial data
- Perform data validation and ensure accuracy, completeness, and compliance with regulatory requirements
- Collaborate with stakeholders to translate regulatory requirements into data-driven outputs and reports
- Utilize SQL and data analysis techniques to retrieve and structure information for decision-making

JUNIOR DATA SCIENTIST APR 2025 – OCT 2025
Absa Group Limited

- Analyzed large-scale transaction datasets using **Hadoop, Hive, SQL, and Python** to identify suspicious patterns and anomalies
- Developed anomaly detection models to flag high-risk financial behavior in financial crime investigations
- Engineered behavioral features from transactional data to improve model performance and interpretability
- Built **Power BI dashboards** to present insights to Financial Crime and Compliance stakeholders
- Delivered data-driven insights to support risk monitoring and strategic decision-making

PERSONAL PROJECTS

Credit Risk & Customer Segmentation Analysis 2026

- Developed a predictive model achieving **0.68 ROC-AUC** and **54% recall on default cases**, improving identification of high-risk customers in an imbalanced dataset
- Engineered key features including credit utilization, payment ratio, and late payment behavior
- Applied clustering techniques to segment customers into risk-based groups for targeted strategies
- Applied best practices in **model evaluation, optimization, and reproducibility**, aligning with MLOps principles
- Built an interactive Power BI dashboard to visualize customer behavior, risk segments, and default trends
- Deployed project outputs via GitHub, demonstrating end-to-end solution development

CERTIFICATION

AWS CLOUD PRACTITIONER
ESSENTIALS
Completion Certification
2025

UDEMY
Data Analysis with Pandas
2025

Country Risk Intelligence Automation System2026

- Designed an automated pipeline using **n8n** and **LLM integration** to generate daily country risk briefings
- Collected and processed news data to identify economic, political, and regulatory risk signals
- Generated structured outputs including **risk scores, alerts, and opportunity indicators**
- Reduced manual monitoring effort by automating end-to-end intelligence generation
- Developed reusable workflows applicable to banking, investment, and risk analysis use cases

Tesla AI Knowledge Assistant | Python · LangChain · FAISS · Llama 3.2 · Streamlit2026

- Built a full Retrieval-Augmented Generation (RAG) pipeline to enable natural language querying of 9 official Tesla documents including annual reports and sustainability reports (2021–2024)
- Implemented semantic search using FAISS vector database and sentence-transformers embeddings for precise document retrieval
- Integrated a locally hosted **Llama 3.2 LLM** via Ollama with zero API dependencies, ensuring full data privacy and no operational cost
- Developed a professional dashboard UI with real-time source citations using Streamlit

REFERENCES

- Available Upon Request